CASE STUDY

MANAGEMENT OF ECLAMPSIA AT 38 WEEKS OF PREGNANCY

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CASE 1: MANAGEMENT OF ECLAMPSIA

Case summary: A pregnant woman at 38 weeks is referred after an eclamptic seizure. Her referral card records magnesium sulfate (MgSO■) 4 g IV and 10 g IM (5 g in each buttock) at 7:00 AM. She has an IV Ringer's lactate line running at a minimal rate. The assessment findings provided include BP 188/106 mmHg, respiratory rate 20/min, clear lungs, pulse 80/min, reflexes 1+, urine protein 2+, fetal movements present and fetal heart rate 144/min. If a catheter is inserted, 140 mL urine is collected. Blood tests including CBC/platelets and liver enzymes are reported normal. The gestational age is confirmed as 38 weeks by LMP, with no signs of labour or other danger signs.

1. Diagnosis

The findings are consistent with **eclampsia with severe-range systolic hypertension** at 38 weeks of pregnancy. Eclampsia is a life-threatening obstetric emergency in which a woman with pre-eclampsia develops seizures. A systolic BP of 188 mmHg requires urgent treatment even though the diastolic BP is 106 mmHg.

2. Step-by-Step Management

Step 1: Immediate assessment and call for help

Call the senior obstetrician, anaesthetist and midwifery/emergency team. Receive the referral and quickly assess airway, breathing and circulation (ABC), level of consciousness and injury. Do not leave the woman alone.

Step 2: Protect the mother during and after a seizure

Place her in the left lateral position, protect her from injury, maintain a clear airway and have suction equipment ready. Give oxygen if she is hypoxic or has respiratory compromise. Do not restrain her and do not put anything in her mouth.

Step 3: Monitor maternal condition

Record BP, pulse, respiratory rate, oxygen saturation, temperature and level of consciousness. Repeat observations frequently, especially while treating severe hypertension and administering MgSO■. The supplied findings are BP 188/106 mmHg, RR 20/min and pulse 80/min.

Step 4: Continue magnesium sulfate safely

The loading dose recorded is 4 g IV plus 10 g IM (5 g in each buttock) at 7:00 AM. Confirm the exact administration time and the protocol used. If following the Pritchard regimen, the first 5 g IM maintenance dose is generally due 4 hours after the loading dose (11:00 AM if the loading dose was at 7:00 AM), provided the woman is clinically safe to receive it. Continue maintenance according to the local protocol. Before each dose, check that the patellar reflex is present, respiratory rate is at least 16/min and urine output is adequate (commonly at least 25–30 mL/hour). Withhold MgSO■ and seek urgent review if toxicity is suspected.

Step 5: Be prepared for magnesium toxicity

Watch for absent knee reflexes, respiratory depression and reduced urine output. Keep calcium gluconate 10% available as the antidote; a commonly used emergency dose is 10 mL IV slowly, administered by a qualified clinician according to local protocol.

Step 6: Control severe blood pressure

Treat persistent severe hypertension urgently using a prescribed first-line medicine such as IV labetalol, IV hydralazine or oral immediate-release nifedipine, following the hospital protocol and checking for contraindications. Recheck BP frequently (often every 15–30 minutes during initial treatment) until it is below the severe range.

Step 7: Insert a urinary catheter and monitor fluids

Insert a Foley catheter using aseptic technique if indicated, measure urine output hourly and maintain a strict fluid balance chart. Record the stated initial collection of 140 mL if this is the volume obtained at insertion. Continue the IV fluid at the prescribed minimal rate and avoid fluid overload because women with severe pre-eclampsia are at risk of pulmonary oedema.

Step 8: Investigations

Send blood for CBC and platelet count, serum creatinine/renal function and liver enzymes. Repeat tests if the clinical condition changes or the senior team requests them. Normal initial results do not remove the need for ongoing monitoring.

Step 9: Assess fetal wellbeing

Confirm fetal movements and assess the fetal heart rate. The reported FHR is 144 beats/min, which is within the usual baseline range. Continue fetal monitoring according to the mother's condition and local obstetric guidance. Maternal resuscitation and stabilization take priority during an acute seizure.

Step 10: Plan delivery after maternal stabilization

At 38 weeks, delivery is indicated after the mother has been stabilized. Inform the obstetrician and neonatal team and prepare for birth. The route depends on maternal and fetal condition, cervical findings and obstetric factors. Induction and vaginal birth may be appropriate when safe; Caesarean birth is used when clinically indicated. Eclampsia alone does not automatically mean Caesarean section.

Step 11: Post-delivery care

Continue MgSO■ for 24 hours after delivery or the last seizure, whichever is later, according to protocol. Continue BP treatment and monitor respiratory rate, reflexes, urine output, consciousness and signs of recurrent seizures or pulmonary oedema. Observe for postpartum haemorrhage and provide support with recovery and breastfeeding when the mother is stable.

3. Medication

Medication	Purpose
Magnesium sulfate (MgSO■)	Prevents recurrent eclamptic seizures.
Labetalol	Controls severe hypertension.
Hydralazine	Alternative medicine for urgent BP control.
Immediate-release nifedipine	Oral option for urgent BP control.
Calcium gluconate 10%	Antidote for magnesium sulfate toxicity.
Ringer's lactate	IV fluid; administer carefully and avoid overload.

Medication safety note

Exact doses and repeat intervals for antihypertensives and MgSO■ must follow the local emergency obstetric protocol and be prescribed/administered by qualified staff. Do not give a maintenance MgSO■ dose unless the required safety checks are satisfactory.

4. Conclusion

This woman has eclampsia at 38 weeks with severe-range systolic blood pressure. The priorities are to protect and stabilize the mother, prevent further seizures with magnesium sulfate, urgently control blood pressure, monitor urine output and laboratory results, assess the baby, and arrange safe delivery. Close monitoring must continue after birth because complications can still occur in the postpartum period.

5. References

• World Health Organization (WHO). WHO recommendations for prevention and treatment of pre-eclampsia and eclampsia.

• National Institute for Health and Care Excellence (NICE). Hypertension in pregnancy: diagnosis and management (NG133).

• Local hospital or national obstetric emergency protocol for magnesium sulfate and severe hypertension management.

Academic note: *This case study is prepared for learning purposes and is not a substitute for senior clinical assessment or the current local treatment protocol.*